

Microsoft Excel

It is an application software or part of MS-office. It is called spread sheet package. The extension name of MS-Excel is .xlsx

Application of MS-Excel: -

It is the accounting package which create the file just like as the ledger of the accounting. It is also known as spreadsheet package. Sheet has a grid of rows and columns which can be used to enter the data. It can perform mathematical or logical calculation and provide their result in anywhere in the work sheet.

Electronic Spreadsheet: -

It is the sheet where we can enter unlimited data in row and column wise which spreading the sheet Here it accepts the data value with having relationship with one data and the related data can be used a formula to produce the result.

How to Start MS-Excel

1. Click on start
2. Choose all Program
3. Choose MS-office
4. Choose Microsoft excel 2007, 2010, 2013, 2016, 2019, 2021, 365

Or

1. Click on the Start
2. Choose Run
3. Type 'excel'
4. Then click ok button

Application Window: -

The outermost window is called application window. It contains title bar, menu bar, status bar, formula bar etc.

Spreadsheet Window: -

The innermost window is called spreadsheet window. It contains only worksheet/spread sheet. In spread sheet window by default opens 3 worksheets. But maximum we can insert 255 worksheets.

Work Book: -

A workbook is a main element of MS-excel. It contains no. of worksheet by defaulting it contain 3 worksheets maximum. It contains 255 worksheets.

Sheet: -

Sheet is an element of workbook. It contains no or rows and no. of columns and also contain row heading and column heading.

Formula bar: -

It is present bellow the formatting toolbar. It has two part, one is formula bar another is name box. The formula bar contains only content of selected cell but the name box contains only address of selected cell.

Scroll bar: -

It is used to move the spreadsheet window up-to-down and left-to-right.

Column: -

The vertical grid lines are called column. In a worksheet there are 256 columns are present according to the 2003 version & according to 2007 version there are 16384 columns present such as A-Z, AA-AZ, BA-BZ, CA-CZ...ZA-ZZ, AAA-AAB,XFD.

The first column heading is called "A" and last column heading is called "XFD".

Row: -

The horizontally grid lines are called row. In a sheet there are 65536 rows. Present according to 2003 version. But according to 2007 and later versions up to Microsoft 365 there is 10,48,576 rows.

Cell: -

The intersection of rows and columns creates a new shape is called cell.

Active Cell: -

The cell contains a black heavy boarder is called active cell.

Fill handle: -

The active cell contains a black button in lower right corner is called fill handle. When you move the mouse pointer over the fill handle then the mouse pointer changes a simple plus (+) sign

1. Anywhere the Work sheet.
2. Over the fill handle.
3. Over the active cell border.

The first cell address is - A1

Last cell of a work sheet is – XFD1048576

Understanding worksheet cell addresses: -

1. Last cell of 1st column - A1048576
2. Last cell of 1st row - XFD1
3. Last cell last row - XFD1048576
4. Last cell last column - XFD1048576
5. First cell of 1st row - A1
6. 1st cell of Last row - A1048576
7. 1st cell of last column - XFD1
8. 1st cell of 1st column - A1
9. Last cell of last column - XFD1048576
10. 1st cell of work sheet - A1
11. Last cell of work sheet - XFD1048576

Shortcut key of Sheet Navigation: -

- | | |
|-------------------------|--|
| 1. (Right arrow) -- | Move the cursor one cell to right. |
| 2. (Left arrow) -- | Move the cursor one cell to left |
| 3. (Top Arrow) -- | Move the cursor one cell up |
| 4. (Down Arrow) -- | Move the cursor one cell down |
| 5. Enter-- | Move the cursor one cell to down |
| 6. Tab-- | Move the cursor one cell to right |
| 7. Shift + tab -- | Move the cursor one cell to left |
| 8. Shift + Enter-- | Move the cursor one cell to up |
| 9. Ctrl + Right Arrow-- | Move the cursor last cell of current row |
| 10. Ctrl + Top Arrow-- | Move the cursor first cell of current column |

- | | |
|-------------------------|---|
| 11. Ctrl + Down Arrow-- | Move the cursor last cell of current column |
| 12. Ctrl + Left Arrow-- | Move the cursor first cell of current column |
| 13. Home-- | Move the cursor first cell of current row |
| 14. Ctrl + Home-- | Move the cursor first cell of the whole work sheet. |
| 15. Page Up-- | One screen to up in vertically order |
| 16. Page Down-- | One screen to down in vertically order |
| 17. Alt-Page up-- | Move one screen left to horizontally order |
| 18. Alt - Page down-- | Move one screen right to horizontally order |
| 19. Ctrl + Page down-- | To select next sheet of the workbook. |
| 20. Ctrl + Page up -- | To select previous sheet of the workbook. |

How to move the cursor instantly in your wanting cell address: -

1. Click on the Name box
2. Type as your cell address
3. Press enters

How to edit cell content: -

1. Click on the cell
2. Click formula bar
3. Using arrow key edit your text
4. Press enters

Or

1. Double clicks on the cell
2. Using arrow key edit your text
3. Press enter ↵

Or

1. Click on the cell
2. Press F2
3. Edit your text using arrow key
4. Press Enter

Save work space: -

Save work books in an arranged work space. When you open a work space file menu Microsoft excel opens each work book saved in the work space.

Web page preview: -

This option is used to see the worksheet in web formula.

Print area: -

This option is used to define what part of worksheet is to be print. This option has two sub-options.

1. Set print area
2. Clear print area

How to set print area to a worksheet: -

- > Choose view menu
- > Choose page break preview
- > Select the area you want to print
- > Click on the file menu
- > Then Choose print Area

-> Set print Area

How to add cells to your existing print area: -

1. On the view menu click page break preview.
2. Select the cell you want to add to print area.
3. Right Click within the selection
4. Click add to print area.

Page Setup: -

This option is used to setup the page margin of the selected print area. When you choose page setup option it opens a dialog box having 4 tabs.

1. Page tab
2. Margin
3. Header & Footer
4. Sheet tab

In page step we can setup the co-orientation of page such as portrait, landscape. In margin tab we can adjust the 4 types of margins and also header space and footer space.

Note: - In the button center of the margin tab you can choose horizontally or vertically to reflect the best position of your data in a page.

Header and Footer: -

The header & footer tab is used to customize the header section or footer section.

How to customize header position: -

1. From the page setup dialog box, click header & footer tab
2. Choose Custom header in left section type appropriate option, in center section write appropriate option, or in right section type appropriate option then click ok
3. Choose custom footer in left section type appropriate option, in center section write appropriate option, or in right section type appropriate option then click ok
4. Finally click ok

Sheet tab: -

In this step we can adjust the printing property of the page and also, we can adjust page order.

Edit menu

Delete: -

This option is used to delete the selected row, column and cell.

Delete Sheet: -

This option is used to delete the selected sheet.

Move or copy Sheet: -

With this option we can copy or move excel work sheet from one work book to another or in same work book.

How to copy a worksheet to another work book: -

1. Open the workbook that you want to copy the work sheet.
2. Open another work book that you want to work sheet copied to
3. Go to workbook with the desired worksheet that you wish to copy. Right click on the tab of the target work sheet
4. Choose move or copy from the edit menu.
5. From the move or copy window appear use to book.

6. Drop down list to select the workbook that you want to copy the worksheet to (ex-book-2)
7. Then select the existing sheet that you want to copy to be placed in front or back. If you selected the sheet-2 the copied work sheet will be placed before the sheet-2
8. Check the “create a copy” check box.
9. Click ok. You will now have. The Same work sheet in both work books.

Note: - To Move a worksheet to another workbook just follow the above steps and simply exclude, checking the created a copy check box.

Fill: -

When you choose edit menu it opens a dropdown list in that list the fill option is very important.

The fill option contains six sub-options

Down: - This option fills the cell content to words down (Ctrl + d)

Step: -

1. Select a cell
2. Type some data
3. Select multiple cells to words down introducing the main cell
4. Click edits
5. Click fills
6. Click down

Right: - This option fills in the cell content to right some step (Ctrl + r)

Up: - Same step

Left: - Same step

Series: -

This option is used to fill the cell content to words row wise or column wise, but it creates a link between tow cells the links are 4 types

1. Linear
2. Growth
3. Data
4. Auto fill

Linear: - This option adds a specially value in each step. That especially value is providing by the user in step box.

Growth: - This option is used to specify multiple value each step.

Date: - This option is used to fill the data in day format, month format and year format.

Auto fill: - It is fills automatically of the selected data in linear wise.

Clear: -

This option is used to clear all the selected text form the selected cell. It has four submenus.

1. all
2. format
3. content
4. comment

All: - This option is used to clear all the contents of selected text and also format of the selected cell.

Format: - This option clear only formats of selected text.

Content: - This option clear only content of selected text not format.

Comment: - This option clear only comment.

How to insert comment: -

1. Click on cell you want to insert comment.
2. Click insert
3. Choose Comment
4. Then Type comment then press enter

Find: -

This option is used to find a particular text with in a sheet or a workbook. To find the text do the following step.

1. Click on edit menu
2. Choose find
3. In find what box type you are finding text.

Within: - This option is used to find the text with in a sheet or work book

Search: - This option is use to search the finding text in row wise, or column wise.

Look-in: - This option is used to find the text only formula cells, only value cells and only comment cells above 3 options. Ex- within, search, looking, choose any one as your choice, then click find next, then click close.

Replace: -

This option is used to a replace a particular word over another word.

Go to: -

This option is used to go move the cursor immediately in this particular cell.

View menu: -

Comment: -

To right anything about the selected cell that is called comments.

Steps

Click on the cell you want to insert comment

1. Choose view.
2. Choose comment.
3. Right click comment
4. Press Enter.

View

In MS-word there are 5 types of views are presented

- 1) Print layout view
- 2) Web layout view
- 3) Normal view
- 4) Out line view
- 5) Pending layout view

But in excel there is only 2 types view is presented

- 1)Normal
- 2)Page break preview

Normal

The normal view display whole sheet of at a time and the default view of ms- excel is normal view

Page break preview

This view display only printed page custom views this option is used to create personal view by a personal name to a selected are.

How to create custom view: -

At first select the area you want to create your view

1. Click view menu
2. Click custom view
3. Click add
4. Type the name of the view
5. Click on ok.

Insert menu: -

Cell: -

Using this option we can insert cell, column rows before a particular position.

How to insert a cell: -

Step: - select the cell click the cell where you want to insert a row cell.

1. Choose insert cell menu
2. Choose cells
3. After clicking the cells, it opens a dialog box containing 4 options
4. Choose any one
5. The option are

1) Shift cell right

Move existing cell to the down and inserts empty cell

2) Shift cell down

Move existing cells to the down and inserts empty cell.

3) Entire row

Move all cells in the row towards down and insert a new row.

4) Entire column

Move all cells in the column to the right and insert row column.

Click on the ok button.

Row

This option is use to insert a blank column row before the selected cell or row.

Column

This option is use to insert a blank column before the selected column or cell.

Work sheet

This option is use to insert a blank work sheet before the selected work sheet.

Name

This option is use to define a name over a particular cell address, in excel every cell have an address the cell address you can use define a name to every cell of the work sheet.

How to define a name: -

1. click on the cell
2. choose insert
3. choose e name
4. choose define
5. choose add
6. type the name of the cell
7. Click ok.

The cell name must be character underscore period and digit.

-> The 1st character of a cell name must be a character the name maximum 253 character.

-> The cell name cannot contain any space.

Comment: -

To right anything about the selected cell is called comment.

How to insert comment: -

- i. Select the cell
- ii. Choose insert
- iii. Choose comment
- iv. Type the comment
- v. Press 'enter'.

Picture: -

Using this opting we can insert different types of pictures, clip art, auto shapes, word arts and organization chart and also customize the above pictures.

Object: -

With this option we can insert different types of objects such as bitmap image, word file, calenderers.

Hyperlink: -

The link bet two different documents is called hyper click by this option; we can create the name bets current document to another document.

How to create hyper link: -

1. Click on the any cell for any text.
2. Choose insert
3. Choose hyper link
4. Crate a document you want to create link
5. Click ok

Format: -

This option plays very important role about the cell navigation.

Cells: -

The cell option is used to format the cells when you click cells option under format menu. It opens format cells dialog box, have contain six types of tabs.

1. Number

The step is used to display the cells number in different category such as number, currency, date, time etc. by defaulters' gene racy.

2. Alignment

The alignment tab is used adjustment of test alignment, text, control, orination.

3. Font

The font tab is use to format the cell such as font size, font style, font type and font color.

4. Border

The pattern tab is use to apply a shading color if the selected cell or cell.

5. Pattern

The pattern tab is used to apply a shading color if the selected cell or cell.

Row: - (Default row height 12.75)

The row option is used to customized selected rows such as, row right, hide, unhide, auto fit.

Column: - (default column width 8.43)

The column option is used to customize the select column such as column right, hide, unhide, auto fit.

Sheet: -

The sheet option is using change sheet name sheet background color, sheet tab color. Cell reference is a one type of criteria trick to which it creates a relationship of every cell address. The cell reference is basically used in formula. When we type any formula in any cell the formula will automatically changes in the next cell. The cell reference is of 3 types.

- i. Relative cell reference
- ii. Absolute cell reference
- iii. Mixed cell reference

1. Relative cell reference

In relative cell reference one reference automatically copies in the next cell of row wise or. Here the cell address automatically changes.

Ex: - A1+B1+C1Ans: B1+C1+D1
A2+B2+C2

2. Absolute cell reference

If the formula with absolute cell reference when you fill the cell it copies the cell formula to the next cell but the formula remains unchanged. CNO change in column or row heading. To create absolute cell reference we can use the '\$' sign, before placing the row heading and column heading.

Ex: - \$A\$1+\$B\$1+\$C\$1Ans: \$A\$1+\$b\$1+\$c\$1
\$A\$1+\$B\$1+\$C41

3. Mixed cell reference

In this cell reference we can create a constant either column heading or row heading.

Ex: - 1. \$A1+\$B1+\$C1Ans: \$A1+\$B1+\$C1
\$A2+\$B2+\$C2` (column constant)
2. A\$1+B\$1+C\$1 Ans: B\$1+C\$1+D\$1
A\$1+B\$1+C\$1 (row constant)

Function: -

It is the another part of MS-excel using function we can solve in easily. The functions are 9 types.

1. Mathematical function:

Abs(), fact(), mod(), pi(), power() product(), sqrt(), sum()

2. Statistical function:

Average(), count(), counta(), count blank(),max(), min(), median()

3. Text function:

Char(), code(), concatenate(), exact(), find() left(), Len(), lower(), upper(), proper(), mid() replace(), rept(), search(), substitute(), trim()

4. **Logical function:**
If(), And(), Or(), Not(), sumif(), Count if()
5. **Date & Time:**
Today(), now(), date(), hour(), minute(), month(), Datedif()Weekday()
6. **Database:**
Dsum(), Dcount(), Dmax(), Dmin(), Daverage()
7. **Look up:**
Lookup Array(), Lookup Vector(), Vlookup(), Hlookup(), match(), Lookup(), offset()
8. **Other Function:**
Age calculation, delta(), dget(), Transpose(), rank(), GCD(), LCM(),
index(),networkdays (), Sum as Running total(),Large (), Small ().
9. **Banking function:**
Fv(), pmt(), ppmt(), ipmt()

1. Mathematical function: -

With this function we can calculate solve some mathematical problem. Here it is essential to know that each function start with equal singe, without equal sign. The functions remain a text of the cell. In mathematical function one is function part another is argument part. The arguments are present inside the function. As an argument we can also give absolute value cell address and cell range.

The cell address is separated by comma sign, but the cell range defined in column (i) sign.

Abs ()

This function writes the absolute value as given argument. In argument you can give the no. or cell address.

=abs (number)

=abs (-5) Ans: -5

= (5) Ans: -5

Fact ()

This function writes the factory number of the given argument.

Syntax: -

=fact (number)

=fact (5, divisor)

=fact (5) Ans: -120

=fact (3) Ans: -6

Mod ()

It is written the reminder part after a divider is divided by a divisor.

Syntax: -

=mod (no, divisor)

Ex: - =mod (10, 4) Ans: -2

Pi ()

The pi () is used to display the value of pi ()

Sqrt ()

This function writ tens the square roots of a given argument.

Syntax: - sqrt (number)

Ex: - sqrt (144) Ans: - 12

Power ()

This power () written the power of given number

Syntax: - =power (base, power)

Ex: - power (5, 3) Ans: - 125

Product ()

This function multiples given number and display the result as an absolute value.

Syntax: - product (no., no., no.)

Ex: - product (4, 2, 10, 5) Ans: - 400

Sum ()

This function adds the valued as an argument and display the result.

Syntax: - sum (no., no., no.....)

Ex: - sum (.)

Syntax: - sum (no., no., no.....)

Ex: -

=SUM (5, 3, 4, 8) Ans: - 20

=sum (A1, B3, C9, G7) Ans: - 44

=SUM (A1:D3) Ans: - 31

=SUM (A1:D3, F1:H3, E7:G8) Ans: - 6

=avg (5, 3, 4, 8) Ans:5

=avg (a1, d3, c9, g7) Ans: 8

=avg (a1:d3) Ans: 7.15

=avg (a1:d3, f1:h3, e7:g8) Ans:5.72

Statistical function ()

It is all so a statistical function, it written the maximum number value as the given argument.

=max (5, 3, 4, 8) Ans: 8

=max (a1, b3, c9, g7) Ans:15

=max (a1:d3, f1:h3, e7:g8) Ans: 20

Min ()

This function writes the numeric minimum number the given argument.

=min (5, 3, 4, 8) Ans:3

Count ()

It is a statistical function it counts the number of cells which contain only Number is the specified range.

Syntax: - =count (range)

=count (a1:d3) Ans:4

Counta ()

This statistical function counts no. of cell that is not empty in the specified range.

Syntax: - =counta (range)

=counta (a1:d3) Ans: 7

Count blank ()

It is a statistical function it counts only the empty cell in the specified range.

Syntax: - =counta blank (range)

=counta blank (a1:a4) =9

=counta blank (a1:a4) =9

Logical function:-

If ():-

The "if ()" checks a condition .if the condition is true, it activate the true part, otherwise it activate the false part .the if condition contain 3 types of expression. The first expression is condition part, the second expression is true part and last expression is false part. Each expression separate with comma (,) sign.

Syntax:- =if (condition, true part, false part)

Nested if ()

Where if contain another if is called "nested if" otherwise it called "larder if". It checks more than one condition at a time .if all the condition are true, it activate the true part, and otherwise it activate the false part.

Syntax:- =if (condition, if (condition, true part, false part) false part)

Or = if (condition, true part, if (condition, true part, false part)

And ()

It is a one type of logical function. It checks two or more condition. If all the condition is true it activate the true part, otherwise activate the false part.

Syntax:- =if (and (cond 1, cond 2, cond 3,), true part, false part)

Obc ()

The logical condition checks more than two conditions at a time if any one condition is true it activate the true part, if all the condition is false it activate the false part.

Syntax:- =if (or (cond-1, cond-2, cond-3,.....), true part, false part)

Not ():-

This logical function checks a condition if the condition is true if activate false part, otherwise, if activate true part.

Syntax:- =if (not (condition), true part, false part)

Text Functions:-

Len ():-

It is a one type of text function if display no. of character as a given argument.

Syntax:- Len ("multimedia") Ans: 10

Left ()

It display the specify no. of character from the left side of given argument.

Syntax:- =left ("text", no. of character)

Ex: - =left ("sikiri", 2) Ans: si

Right ():-

It is written the specify no. of character from the right side of given string

Syntax:-

Right [text no. of character]

Ex = Right [sikiri, 2] Ans: Ri

Mid ():-

It writes the specified no. of character from the middle position of the given string.

Syntax:-

Mid (Text, starting no. of character)

Ex= mid (computer 4, 5)-puter

Lower:-

It is a text function which converts all the, character in a given string to lower case.

Syntax = Lower (string)

Ex – Lower (LILI) - lili

Upper ()-

It is a text function which converts all the characters in a given string to upper case.

Syntax – upper (string)

Ex –= upper (lili) Ans:LILI

Proper ()

It is a text function is convert the first character of each word in a given string to upper case and rest will be lower case.

Syntax: - Proper (“text”)

Ex: - =Proper (KODALA) Ans: Kodala

Concatenate ()

It is text function which is use to joint no. of string in to a single string.

Syntax: - =concatenate (“string”, “string”)

=concatenate (“kodala”, “ganjam”) Ans:kodalaganjam

Char ()

It display the character specified the given ASCII cod.

Syntax: - =char (number)

=char (65) Ans:A

Code ()

It is written the numeric code from the 1st character of the given string.

Syntax: - =code (“text”)

=code (t) Ans:43

Exact ()

It checks two string if both string are saying it written the true otherwise it written the false.

Syntax: - =exact (“text-1”, “text-2”)

=exact (kodala, kodala) Ans:true

=exact (kodala, KODALA) Ans:false

Replace ()

This option replaces a part of text string with a different text string.

Syntax: - =replace (add text, start no., no. of character, new text)

=replace (“back”, 2, 2, “oo”) Ans:book

Rept ()

This option reports in the text string the no. of time as a given argument.

Syntax: - =rept (“text”, no. of time)

=rept (“lili”, 2) Ans:lili, lili

Find ()

This button is use to find a text with if another text.

Syntax: - =find (“find text”, “with in text”, string no.)

=find (“I”, “sikiri”) Ans:2

Search ()

This option is use to search of text with in another text.

Syntax: - =search ("search text", "with in text", starting no.)

=search (sikiri) Ans: 2

Substitute ()

This option replaces existing text with a new text in a text string.

Syntax: - =substitute ("text", "old text", "new text")

Ex =Substitute ("back","ac","ov") book

TRIM:-

This option is use to remove all spaces form a text and term except of single spaces between words.

Date and time function:-

Day ()

It writes the day of the month according to serial No, 1to31.

Syntax: - = day (No "Serial No.")

Ex = Day ("15 may") 15

Hour ()

It writes the hour as a no. from 0 to 23

Syntax: - =hour ("serial No")

Ex =hour ("3.30; 3 am")=3

Minute ()

It returns the minute of corresponding serial no.

Syntax: - =minute (sl no.)

Ex = ("5:50"am) 50

Today ()

It writes current date of the system.

Ex = Today () 10/25/2011

Now ()

It display the current date and time of the system.

Ex = Now () 10/25 /2011, 1:25:54

Weekday ()

It writes the day of the week corresponding to serial oc.

Syntax =weekday (mm/dd/yy)

Ex =week day (19/24/2011)2

Auto formatting:-

Some predefined formats are present in the system is called Auto format. Using this option, we can use the auto formats to our selected cells of workbook window.

Conditional formatting:-

This option is use to customize the selected cells provide some condition. So it is called conditional formatting.

-----More notes about the functions are studied in classes-----